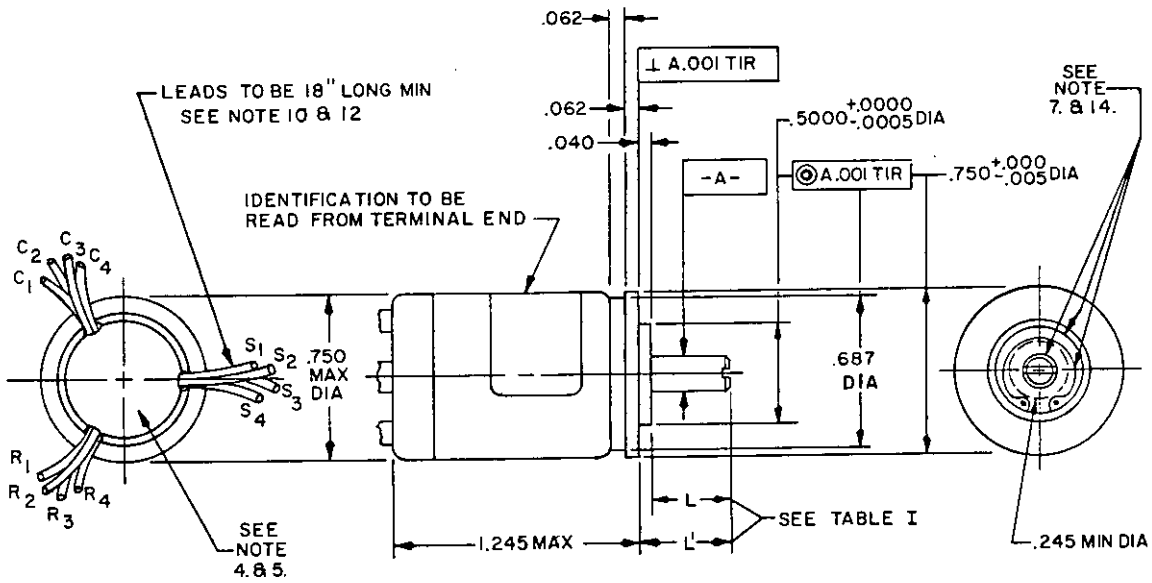
MILITARY SPECIFICATION SHEET

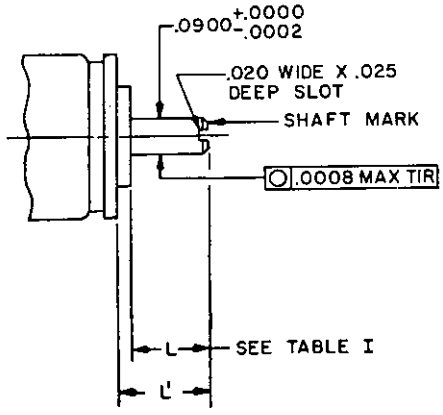
RESOLVER, ELECTRICAL

TYPE 08R21W4B

The complete requirements for procuring the resolver described herein shall consist of this document and the latest issue of MIL-R-23417.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.





FSC 5990

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NOTES:

1. Dimensions are in inches.
2. Unless otherwise specified, tolerances on decimals are ±.005 and angles ±2°.
3. Perpendicularity and concentricity shall be measured with the unit in a vertical position supported by the shaft, then the housing rotated.
4. Terminal lead wires shall depart through the back surface of the terminal end of the resolver in any acceptable manner.
5. Minor variations of unit configuration are permitted for undimensioned design detail.
6. T.R. variation and phase shift variation (rotor/compensator) will be tested as follows:
   Qualification test - Variation shall remain within specification limits when subjected to individual or simultaneous variations of voltage (.8 to 10 volts), frequency (380 to 420 Hz) and temperature (-55°C to +125 °C). Test by applying input voltages of .8, 5 and 10 volts with frequency and temperature extremes simultaneously applied.
   Acceptance test - Test by applying input voltages of .8, 5 and 10 volts at rated frequency and room ambient temperature.
7. Mark on housing is to match shaft mark within 10° when rotor is set at approx. resolver zero, and will bear no relation to the center lines shown on the drawing.
8. Pending issuance of a qualified products list (QPL) a sample of the resolvers to be furnished under this specification shall be subjected to the preproduction tests as specified in MIL-R-23417.
9. For information only see synchro mounting clamp MS17183 or equal.
10. Leads shall be type E-28.
11. A voltage value of .5 to 26V shall be entered in the 'Volts' designation space on the identification plate.
12. The weight shall be attached to the end of each wire, pulling straight away from the resolver. The lead shall then be bent 90° at the point of exit from the housing and while in this attitude the resolver rotated 360° counterclockwise. Any indication of the lead pulling loose, wire strands breaking, or permanent deformation of the insulation shall be cause for rejection under qualification tests.
13. With the resolver energized under standard test conditions during qualification the resonant frequency will be recorded during qualification as that point where the output voltage has a maximum value and phase shift -90°. Low corner frequency (referenced as design information) is the point where output voltage is down 3db and phase shift 45°.
14. The front retainer shall be threaded or a snap ring.

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RESOLVER, ELECTRICAL: TYPE 08R21W4B

PERFORMANCE REQUIREMENTS AND CHARACTERISTICS

Requirement	Value	Unit	Tolerance
1. Excitation winding	Stator	---	---
2. Rated frequency	400	Hz	±1%
3. Input voltage:			
Maximum	15	Volts rms	±1%
Test	10	Volts rms	±1%
Minimum	.5	Volts rms	±1%
4. Primary current @ test voltage	5	Milliamps	Nominal
5. Primary power	.015	Watts	Nominal
6. DC resistance:			
Rotor	500	Ohms	Reference
Stator	555	Ohms	Reference
Compensator	350	Ohms	Reference
7. Impedance:			
Zro	570 +j2215	Ohms	R±15%, +jx±10%
Zso	615 +j1930	Ohms	R±15%, +jx±10%
Zrs	995 +j350	Ohms	Reference
Zco	385 +j525	Ohms	R±15%, +jx±10%
Zcs	492 +j59	Ohms	Reference
8. Resonant frequency (see note 13):			
Rotor	60	kHz	Minimum
Compensator	400	kHz	Minimum
9. Low corner frequency (see note 13)	100	Hz	Reference
10. Transformation ratio:			
Output/input	.970	---	±2%
Rotor/compensator	1.96	---	±2%
11. T.R. variation Rotor/compensator (see note 6) (with test voltage as reference) @ .8 to 10 volts	±.15	Percent	Maximum
12. Equality of transformation ratio:			
Output/input	.1	Percent	Maximum
Rotor/compensator	.1	Percent	Maximum
13. Phase shift:			
Output/input	15.5	Degrees	±2
Rotor/compensator	0	Degrees	±0.15
14. Phase shift variation (Rotor/compensator) (see note 6) Due to voltage @ .8 to 10 volts	±8	Minutes	Maximum
15. Interaxis error	±5	Minutes	Maximum
16. Functional error	±.1	Percent	Maximum
17. Null voltage:			
Fundamental	10	Mv rms	Maximum
Total	12	Mv rms	Maximum
18. Shift of resolver zero:			
With voltage	±1.5	Minutes	Maximum
With frequency	±.5	Minutes	Maximum
19. Harmonic distortion	.015	Percent	Maximum
20. Friction torque:			
@ 23° C	.04	Oz-In	Maximum
@ -55° C and +125° C	0.1	Oz-In	Maximum

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Requirement	Value	Unit	Tolerance
21. Radial play	.0004 (For reversal of 1/4 lb load)	Inches	Maximum
22. End play	.0001-.0015 (For reversal of 1/2 lb load)	Inches	Min-Max
23. Wire lead stress (see note 12)	2	Pound	Minimum

Part number: The part number consists of the basic number of this specification sheet and a dash number taken from table I.

Example:

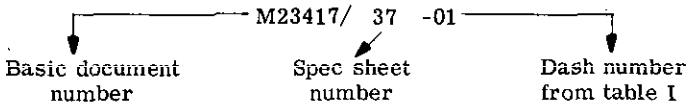


TABLE I. Dash numbers and characteristics.

Dash No.	Shaft diameter	Shaft style	Termination	Shaft length	
				(L (ref)	L' - dim to first mounting flange surface (±.010)
-01	.090	Plain	Wire leads	.325	.365
-02	.090	Plain	Wire leads	.406	.446
-03	.090	Plain	Wire leads	.500	.540

Custodians:
Army - MU
Navy - AS
Air Force - 80

Review activities:
Army - MU, EL
Navy - AS, SH, EC
Air Force - 11, 17, 80
DSA - ES

User activities:
Army - MI, ME, AV
Navy - CG, SH
Air Force - 19

Preparing activity:
Air Force - 80

Agent:
DSA - ES

(Project 5990-0253)

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